kimi-k3-hf-model-card

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tags:

- compressed-tensors
- conversational license: other license_name: "kimi-k3" library_name: transformers pipeline_tag: image-text-to-text

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1. Model Introduction

Kimi K3 is an open-weight, native multimodal agentic model and our most capable model to date. It is a 2.8T-parameter model built on Kimi Delta Attention (KDA) and Attention Residuals (AttnRes), with native vision capabilities and a 1-million-token context window. It is the world's first open 3T-class model, designed for frontier intelligence across long-horizon coding, knowledge work, and reasoning.

Key Features

- **New Architecture**: Kimi K3 is built on Kimi Delta Attention (KDA) and Attention Residuals (AttnRes), and scales up MoE sparsity with a Stable LatentMoE framework that activates 16 out of 896 experts — yielding an approximate 2.5× improvement in overall scaling efficiency over Kimi K2.
- **Long-Horizon Coding**: Operating with minimal human oversight, Kimi K3 sustains long engineering sessions, navigates massive repositories, and orchestrates terminal tools — from GPU kernel optimization and compiler development to vision-in-the-loop game dev, CAD, and even chip design.
- **Agentic Knowledge Work**: Kimi K3 advances end-to-end knowledge work, producing deep research with interactive visualizations, widgets and dashboards, and motion design and video editing, powered by its native multimodal architecture.
- **Native Multimodality & Long Context**: Kimi K3 understands text, images, and video within the same model, and supports a 1-million-token context window.
- **Open Frontier Weights**: We release the full Kimi K3 model weights under the Kimi K3 License, making frontier intelligence openly available for research, deployment, and further innovation.

2. Model Summary

Architecture	Mixture-of-Experts (MoE)
Total Parameters	2.8T
Activated Parameters	104B
Number of Layers	93
Number of Dense Layers	1

Attention-Layer Composition	69 KDA + 24 Gated MLA
Attention Hidden Dimension	7168
Number of Attention Heads	96
Latent MoE Dimension	3584
MoE Hidden Dimension (per Expert)	3072
Number of Experts	896
Selected Experts per Token	16
Number of Shared Experts	2
Vocabulary Size	160K
Context Length	1048576
Attention Mechanism	KDA & Gated MLA
Activation Function	SiTU-GLU
Vision Encoder	MoonViT-V2
Parameters of Vision Encoder	401M
Quantization	MXFP4 weights / MXFP8 activations (quantization-aware training)
Modality	Text, Image

3. Evaluation Results

Benchmark	**Kimi K3** **(max)**	**Claude Fable 5** **(max, w/ fallback)**	**GPT-5.6 Sol** **(max)**	**Claude Opus 4.8** **(max)**	**GPT-5.5** **(xhigh)**	**GLM-5.2** **(max)**
Reasoning & Knowledge						
GPQA Diamond	93.5	92.6	94.1	91.0	93.5	91.2
CritPt	23.4	28.6	32.3	20.9	27.1	20.9
AA-LCR	74.7	70.0	73.7	67.7	74.3	71.3
HLE-Full	43.5 / 56.0	53.3 / 63.0	44.5 / 58.0	49.8 / 57.9	41.4 / 52.2	—
Coding						
DeepSWE	67.5	70.0	73.0	59.0	67.0	46.2
ProgramBench	77.8	76.8	77.6	71.9	70.8	63.7
Terminal-Bench 2.1	88.3	88.0	88.8	84.6	83.4	82.7
FrontierSWE	81.2	86.6	71.3	66.7	64.9	67.3
SWE-Marathon	42.0	35.0	39.0	40.0	14.0	13.0
PostTrainBench	36.6	41.4	34.6	34.1	28.4	34.3
MLS-Bench-Lite	48.3	49.9	46.2	42.8	35.5	40.4
SciCode	58.7	60.2	56.1	53.5	56.1	50.5
Kimi Code Bench 2.0	72.9	76.9	64.8	71.7	69.0	64.2
Agentic						
BrowseComp	91.2	88.0	90.4	84.3	84.4	—
DeepSearchQA (F1)	95.0	94.2	—	93.1	—	—
ResearchRubrics	76.2	—	73.8	73.5	64.0	71.1
GDPval-AA v2 (Elo)	1686	1747	1736	1593	1491	1510
Toolathlon-Verified	76.5	77.9	74.9	76.2	73.5	59.9
MCPMark-Verified	94.5	87.4	92.9	76.4	92.9	—

Benchmark	Kimi K3 (max)	Claude Fable 5 (max, w/ fallback)	GPT-5.6 Sol (max)	Claude Opus 4.8 (max)	GPT-5.5 (xhigh)	GLM-5.2 (max)
MCP-Atlas	84.2	84.7	83.6	83.6	82.8	82.6
AutomationBench	30.8	29.1	29.7	27.2	22.7	12.9
JobBench	54.3	57.4	45.4	48.4	38.3	43.4
AA-Briefcase (Elo)	1548	1583	1495	1354	1158	1260
Agents' Last Exam	28.3	25.7†	29.6	27.0	26.6	20.4
APEX-Agents	41.0	43.3	39.9	39.4	38.5	35.6
OfficeQA Pro	63.3	69.9	63.2	63.9	60.9	41.4
SpreadsheetBench 2	34.8	34.7	32.4	31.6	29.1	28.1
OSWorld-Verified	84.8	85.0	83.0	83.4	79.0	—
OSWorld 2.0	58.3	66.1	62.6	55.7	49.5	—
SaaS-Bench	60.1	—	61.4	56.1	43.8	—
τ^3-Banking	33.4	26.8	33.0	27.6	31.3	26.8
Harvey Lab-AA	94.6	93.6	87.2	91.1	86.3	91.0
CorpFin v2	71.6	71.8	64.4	66.7	68.4	66.1
Finance Agent v2	54.4	56.3	53.8	53.9	51.8	49.7
Legal Research Bench	44.2	49.5	48.1	43.8	40.4	31.3
Vision						
WorldVQA ForceAnswer	51.0	56.7	41.8	39.1	38.5	—
OmniDocBench	91.1	89.8	85.8	87.9	89.4	—
PerceptionBench	58.5	57.2	59.7	47.2	55.8	—
Video-MME (w. sub)	90.0	—	89.5	86.0	89.3	—
MMVU	82.1	—	81.2	79.2	81.7	—
BabyVision w/ python	85.7	90.5	88.9	81.2	83.6	—
MMMU-Pro	81.6 / 83.4	81.2 / 86.5	83.0 / 84.6	78.9 / 82.7	81.2 / 83.2	—
CharXiv (RQ)	84.8 / 91.3	88.9 / 93.5	84.6 / 89.1	80.5 / 89.9	84.1 / 89.0	—
MathVision	94.3 / 97.8	94.8 / 98.6	95.8 / 97.8	86.7 / 97.1	92.2 / 96.8	—
ZeroBench (pass@5)	23.0 / 41.0	23.0 / 46.0	17.0 / 35.0	17.0 / 34.0	22.0 / 41.0	—

▶ **Footnotes**

4. Native MXFP4 Quantization

Kimi K3 applies quantization-aware training from the SFT stage onward, using MXFP4 weights with MXFP8 activations for broad hardware compatibility.

5. Deployment

> [!Note] You can access Kimi K3's API on https://platform.kimi.ai by selecting `kimi-k3`, and we provide OpenAI/Anthropic-compatible API for you. Currently, Kimi K3 is recommended to run on the following inference engines:

- vLLM — see recipes
- SGLang — see cookbook
- TokenSpeed — see recipes

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6. Model Usage

Kimi K3 always has thinking enabled, and will return `reasoning_content`. Thinking effort is configured with the top-level `reasoning_effort` request field, which supports `"low"`, `"high"`, and `"max"` (default `"max"`).

Kimi K3 was trained in the preserved thinking history mode. For multi-turn conversations and tool calls, Kimi K3 requires the complete assistant message returned by the API to be passed back to `messages` as-is — including `reasoning_content` and `tool_calls`, not just `content`:

```
import openai

def chat_with_preserved_thinking(client: openai.OpenAI, model_name: str):
    messages = [
        {
            "role": "user",
            "content": "Tell me three random numbers."
        },
        {
            "role": "assistant",
            "reasoning_content": "I'll start by listing five numbers: 473, 921, 235, 215,
        222, and I'll tell you the first three.",
            "content": "473, 921, 235"
        },
        {
            "role": "user",
            "content": "What are the other two numbers you have in mind?"
        }
    ]

    response = client.chat.completions.create(
        model=model_name,
        messages=messages,
        stream=False,
        max_tokens=4096,
        reasoning_effort="max",
    )
    # the assistant should mention 215 and 222 that appear in the prior reasoning content
    print(f"response: {response.choices[0].message.reasoning}")
    return response.choices[0].message.content
```

For full guides and examples (vision input, structured output, partial mode, tool choice, dynamic tool loading, context caching), see the Kimi K3 Quickstart and Thinking Effort.

Coding Agent Framework

Kimi K3 works best with Kimi Code CLI as its agent framework. We warmly invite you to give it a try — run Kimi Code in your terminal and select Kimi K3 using the `/model` command. We hope you enjoy building with Kimi K3, and we would love to hear your feedback!

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7. License

Both the code repository and the model weights are released under the Kimi K3 License.

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8. Contact Us

If you have any questions, please reach out at support@moonshot.ai.